DATA SCIENCE PROJECT REPORT

Titanic Survival Prediction

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1. Project Title

Titanic Survival Prediction using Machine Learning

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2. Project Description

This project focuses on analyzing the Titanic dataset and building a machine learning model to predict whether a passenger survived or not. It covers the complete data science process including data preprocessing, visualization, feature engineering, and prediction.

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3. Objective

The objective of this project is to understand how data science techniques can be applied to a real-world dataset to analyze data and make predictions.

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4. Learning Outcomes & Skills

- Data Cleaning and Preprocessing using Pandas and NumPy
- Data Visualization using Matplotlib and Seaborn
- Feature Engineering
- Machine Learning Model Building
- Model Evaluation using Accuracy and Confusion Matrix

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5. Dataset Used

The dataset used is the Titanic dataset, which contains passenger information such as:

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

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6. Steps Followed

6.1 Data Collection

The dataset was loaded into Google Colab using a CSV file.

6.2 Data Cleaning

- Missing values in the Age column were filled using median
- Missing values in the Embarked column were filled using mode
- Unnecessary columns like Name, Ticket, Cabin, PassengerId were removed

6.3 Data Visualization

The following visualizations were created:

- Survival Count Plot
- Age Distribution Plot
- Correlation Heatmap

6.4 Feature Engineering

- Created a new feature:
  FamilySize = SibSp + Parch + 1
- Converted categorical data:
  - Male → 0
  - Female → 1
- Applied one-hot encoding on Embarked column

6.5 Model Building

- Used Logistic Regression for classification
- Split data into training and testing sets (80:20)
- Trained the model using training data
- Predicted results using test data

6.6 Model Evaluation

- Evaluated model using Accuracy Score
- Used Confusion Matrix to check predictions

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7. Features Used

- Age
- Sex
- Pclass
- Fare
- FamilySize
- Embarked

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8. Machine Learning Model

Logistic Regression was used because it is suitable for binary classification problems like predicting survival (Yes/No).

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9. Results

The model achieved an accuracy of approximately **75%**, which indicates good performance.

Sample Output:

Accuracy: 0.75

Confusion Matrix:

[82 24]

[19 54]

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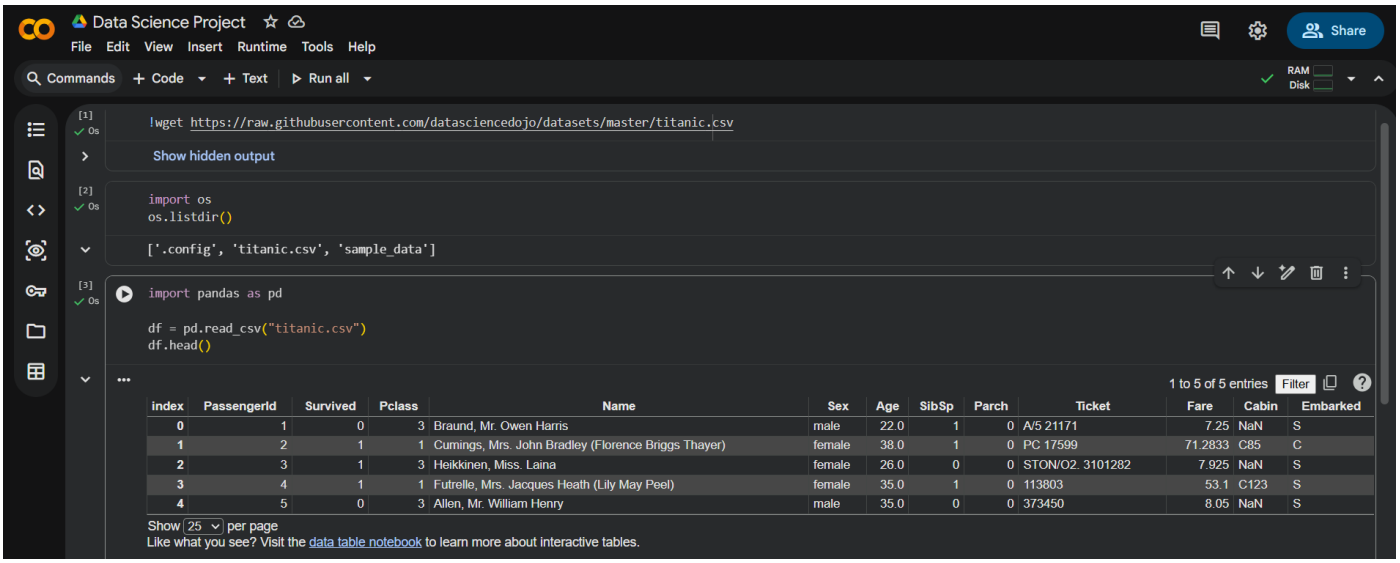
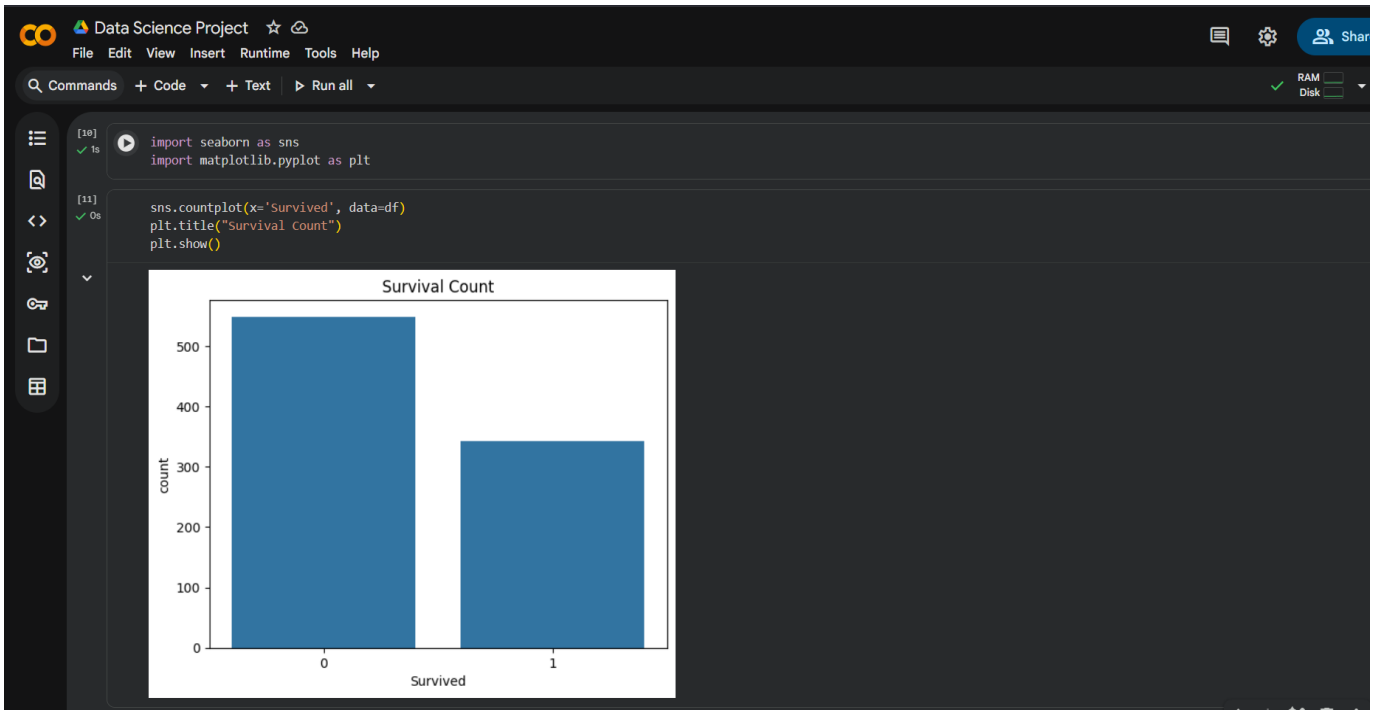
10. Screenshots

1. Dataset preview (df.head())

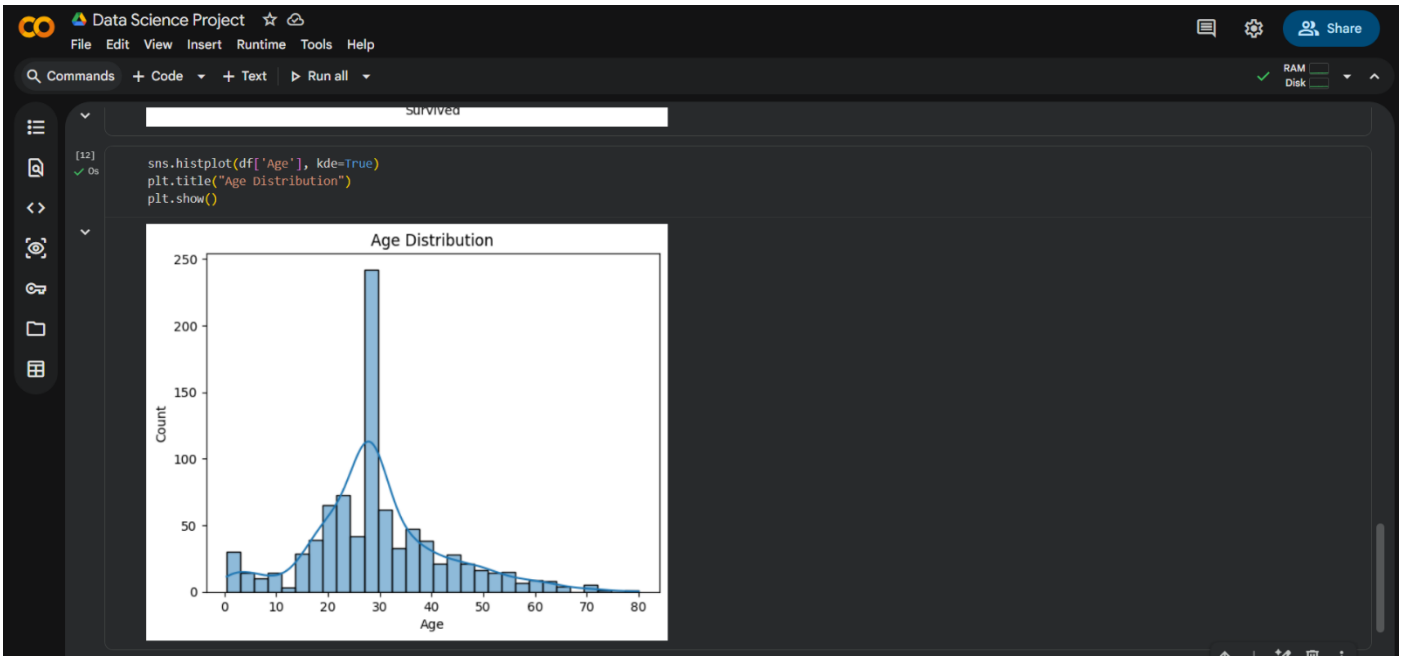


index	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.925	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05	NaN	S

2. Survival Count Graph

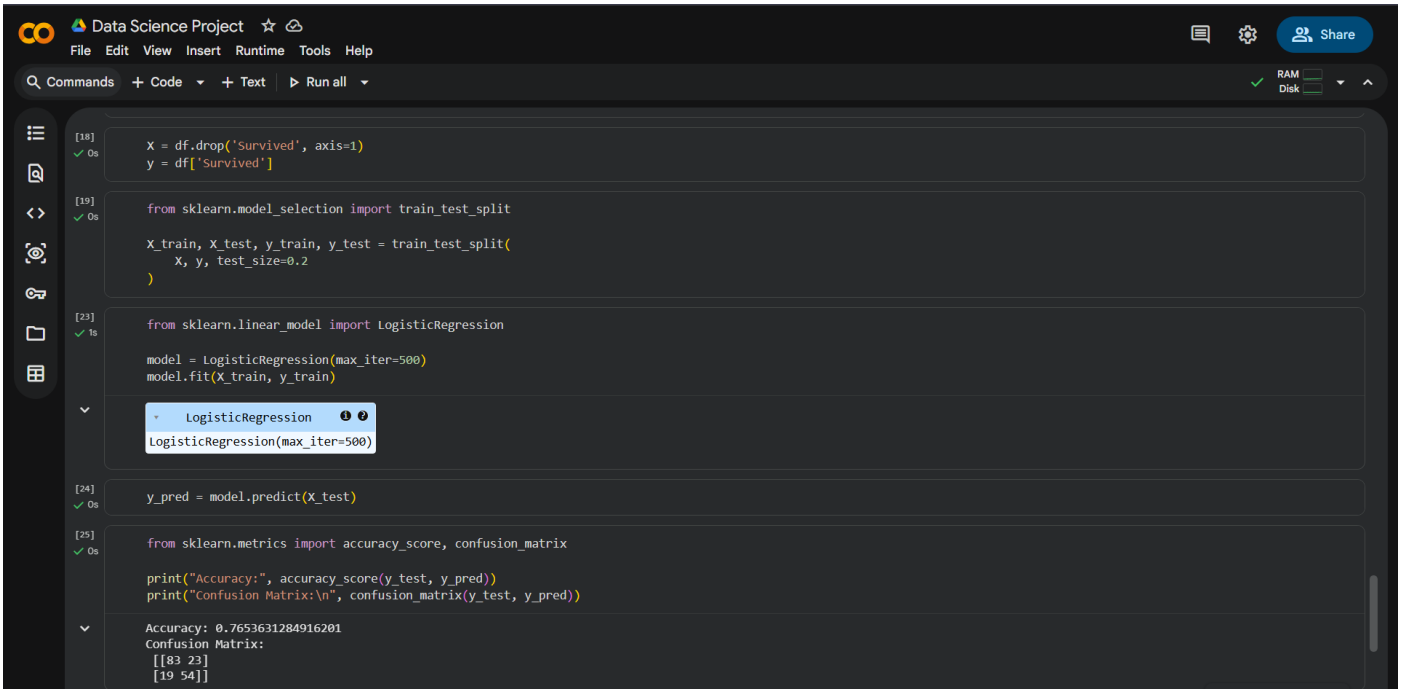


3. Age Distribution Graph



4. Correlation Heatmap

5. Accuracy and Confusion Matrix



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11. Conclusion

This project demonstrates how data science techniques can be used to analyze data and make predictions. By cleaning data, visualizing patterns, and applying a machine learning model, we successfully predicted passenger survival. This project provides a clear understanding of the data science workflow.